

KENYATTA UNIVERSITY
DEPARTMENT OF MATHEMATICS

ECU202: ODE 1 CAT 1

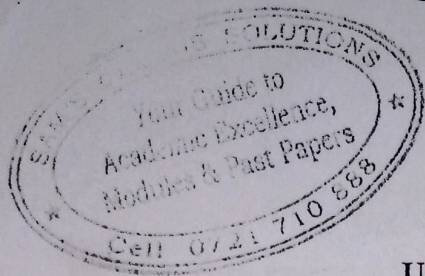
TIME: 1 HOUR

17/11/2021

Answer ALL the Questions.

- 1) Find the differential equation of the family of curves $y = ae^{2x} + be^{3x}$ where a and b are parameters. (3 marks)
- 2) Determine C_1 and C_2 so that $y(x) = C_1 \sin 2x + C_2 \cos 2x + 1$ will satisfy the condition $y\left(\frac{\pi}{8}\right) = 0$ and $y'\left(\frac{\pi}{8}\right) = \sqrt{2}$ (4 marks)
- 3) Show that $3x^2y^2dx + (2x^3y + 4y^3)dy = 0$ is an exact differential equation, - Exact D.E. hence solve it. (7 marks)
- 4) Solve $x \frac{dy}{dx} + \cos y = 0$, given $y(\sqrt{2}) = \frac{\pi}{4}$. (5 marks)
- 5) Solve $\frac{dy}{dx} - \frac{5}{x}y = -5x^2$ - 1st order linear DE (5 marks)
- 6) Solve $y' - \frac{3y}{x} = x^4y^{\frac{1}{3}}$ (6 marks)

"No problem can be solved from the same level of consciousness that created it."
Albert Einstein



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017 SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ENGINEERING)

ECU 202: DIFFERENTIAL EQUATIONS FOR ENGINEERS

DATE: Friday, 9th June 2017

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS: Answer Question ONE and any other TWO questions

QUESTION ONE (30 MARKS)

- a) Determine the order, degree and linearity of the differential equation $\sqrt{y'''} + 2y' = x$ (3 marks)
- b) Form the differential equation for $y = a \cos 3x + b \sin 3x$ where a and b are arbitrary constants (3 marks)
- c) Solve the differential equation $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$ (3 marks)
- d) If $x^k y^m$ is an integrating factor for $\frac{dy}{dx} = \frac{-y + x^4 y^2}{x}$, find the values of k and m . (5 marks)
- e) The population of a certain country grows at a rate proportional to the population at any time. If the population triples in 30 years, in how many years will it double? If the initial population was 5 million people, how long would it take to be 23 million (5 marks)
- f) Use the method of undetermined coefficients to solve the differential equation $(D^2 - 6D + 25)y = 75x^2 - 136x + 80$ (5 marks)
- g) Use the variation of parameters method to solve the differential equation $y'' - 2y' + y = x^{-2}e^x$. (6 marks)

QUESTION TWO (20 MARKS)

- a) By eliminating the arbitrary constants find the differential equation associated to the equation
 $y = e^{-4x} (c_1 \cos 5x + c_2 \sin 5x)$ (3 marks)

- b) For what values of k if any is $y = ce^{kx}$ a solution to the differential equation
 $y''' + 2y'' + 9y' - 20y = 0$. (5 marks)

- c) Explain what is meant by the Wronskian of the terms u_1, u_2 , and u_3 , hence use the  Wronskian to show that if the solutions of a differential equation are e^{3x} and xe^{3x} , then they are linearly independent. Give the general solution of the differential equation. (5 marks)

- d) Solve the differential equation $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} = \cos x$, $y(0) = y'(0) = 1$ (7 marks)

QUESTION THREE (20 MARKS)

- a) Solve the differential equation $(x^2 + y^2) dx = 2xy dy$ (6 marks)

- b) Show that the differential equation $(ye^{xy} + 2x) dx + (xe^{xy} - 3y + 5) dy = 0$ is exact hence solve it
 $\frac{\partial^2}{\partial x^2} + x \frac{\partial^2}{\partial x \partial y}$ (6 marks)

- c) Solve $(x^2 + 1)y'' + xy' - 4y = 0$. (8 marks)

QUESTION FOUR (20 MARKS)

- a) The total cost of production y and the level of output x are related to the marginal cost of production by the equation $(6x^2 + 2y^2) dx - (x^2 + 4xy) dy = 0$. What is the relation between total cost and output, if $y = 2$ when $x = 1$? (6 marks)

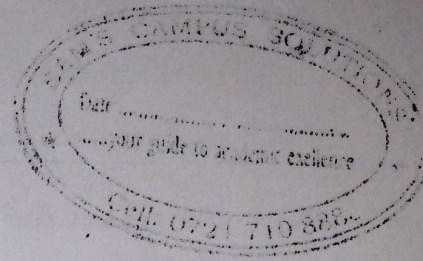
- b) Solve $(1 - x^2) \frac{dy}{dx} - xy = 1$? (5 marks)

- c) A body at a temperature of 50°F is placed outdoors where the temperature is 100°F . If after 5 minutes the temperature of the body is 60°F , find

- i) How long it will take the body to reach a temperature of 75°F . (6 marks)
 ii) The temperature of the body after 20 minutes. (3 marks)



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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELORS OF
SCIENCE IN ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, BIOMEDICAL
AND BIOSYSTEMS ENGINEERING

ECU 202: ENGINEERING MATHEMATICS VII

DATE: MONDAY 13TH APRIL 2015

TIME: 11.00 A.M. – 1.00 P.M.

INSTRUCTIONS

Answer Question ONE and any other Two Questions

QUESTION 1: (30 MARKS)

- (a) Show $y = C_1 e^{-2x} + C_2 x e^{-2x} + x^2 + 3x + 1$ satisfies the differential equation $y'' + 4y' + 4y = 4x^2 + 20x + 18$. (5 marks)
- (b) Show that y^4 is an integrating factor for the differential equation $2xy^{-1}dx + (y^{-3} + 3x^2y^{-2})dy = 0$. Solve it giving the answer in the form $ax^2y^3 + by^2 = C$. (5 marks)
- (c) Show that the differential equation $\frac{dy}{dx} = \frac{2y^2 - x^2}{2xy}$ is homogeneous and that the solution is $y^2 = x^2 \ln|Cx^{-1}|$. (5 marks)
- (d) A metallic bar at 280°F is placed in a room at 35°F . The temperature of the bar after 20 minutes is 200°F . Find an expression for the temperature of the bar at any time t . How long does it take the bar to attain a temperature of 88°F . (5 marks)
- (e) Find the general solution to the differential equation $(D^4 - 6D^3 + 10D^2 + 2D - 15)y = 0$. (5 marks)
- (f) Use the method of **undetermined coefficients** to solve the differential equation $y'' + 4y' = 16 \sin 2x$. (5 marks)

QUESTION 2: (20 MARKS)

- (a) Find the differential equation associated to the primitive $\ln(xy) = Ax + B$ (5 marks)
- (b) Determine the values of m and k for which $x^m y^k$ is an integrating factor for the differential equation $(8x^3 y + 15y^3)dx + (4x^4 + 18xy^2)dy = 0$. Solve the de. (7 marks)
- (c) Solve the Bernoulli equation $\frac{dy}{dx} - \frac{3}{x}y = x^4 \sqrt[3]{y}$ if $y(1) = -1$. Give your answer in the form $y = f(x)$. (8 marks)

QUESTION 3: (20 MARKS)

- (a) Find the orthogonal trajectories to the family of curves $y(x^2 + 1) = Cx$ (6 marks)
- (b) Given that $y = e^{-x} \sin x$ is one solution to the differential equation $(D^4 - 6D^3 + 11D^2 + 34D + 50)y = 0$, find the general solution. (7 marks)
- (c) Show that the transformation $z = \ln y$ transforms the differential equation $\frac{dy}{dx} + \frac{2}{x}y \ln y = 4xy$ into the form $\frac{dz}{dx} + \frac{2}{x}z = 4x$ hence solve the differential equation. (7 marks)

QUESTION 4: (20 MARKS)

- a) Show that the differential equation $x^3 \frac{d^3 y}{dx^3} - 6x \frac{dy}{dx} + 12y = 0$ has three linearly independent solutions of the form $y = x^r$ and find the general solution. (6 marks)
- (b) Consider the homogeneous linear differential equation of order two given by $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 2y = 0$. Use the transformation $x = e^t$ to eliminate x and its derivatives from this equation, hence solve (7 marks)
- (c) If $(x^2 + y^2)dx - 2xydy = 0$, show that its solution is $y^2 = x^2 + cx$. (7 marks)



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING),
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING),
BACHELOR OF SCIENCE (CIVIL ENGINEERING),
BACHELOR OF SCIENCE (PETROLEUM ENGINEERING) AND
BACHELOR OF SCIENCE (BIOMEDICAL AND BIOSYSTEMS ENGINEERING)
ECU 202: DIFFERENTIAL EQUATIONS FOR ENGINEERS

1ST SEMESTER 2021/2022

TIME: 2 HOURS

INSTRUCTIONS: Answer Question **One** and any other **TWO** questions

QUESTION ONE (30 MARKS)

- a) Determine the order, degree and linearity of the differential equation $\sqrt{y'''} + 2y' = x$ (3 marks)
- b) Form the differential equation for $y = a \cos 3x + b \sin 3x$ where a and b are arbitrary constants (3 marks)
- c) Solve the differential equation $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$ (3 marks)
- d) If $x^k y^m$ is an integrating factor for $\frac{dy}{dx} = \frac{-y + x^4 y^2}{x}$, find the values of k and m hence solve the differential equation. (5 marks)
- e) The population of a certain country grows at a rate proportional to the population at any time. If the population triples in 30 years, in how many years will it double? If the initial population was 5million people, how long would it take to be 23million (5 marks)
- f) Use the method of undetermined coefficients to solve the differential equation $(D^2 - 6D + 25)y = 75x^2 - 136x + 80$ (5 marks)
- g) Use the variation of parameters method to solve the differential equation $y''' - 2y' + y = x^{-2}e^x$. (6 marks)

$$35e^{-4x} \cos 5x - 35e^{-4x} (2 \cos 5x - 13e^{-4x} \cos 5x - 13e^{-4x} \sin 5x)$$

QUESTION TWO (20 MARKS)

a) By eliminating the arbitrary constants find the differential equation associated to the equation

$$y = e^{-4x} (c_1 \cos 5x + c_2 \sin 5x)$$

(3 marks)

b) For what values of k if any is $y = ce^{kx}$ a solution to the differential equation

$$y''' + 2y'' + 9y' - 20y = 0$$

(5 marks)

c) Explain what is meant by the Wronskian of the terms u_1, u_2 , and u_3 , hence use the Wronskian to show that if the solutions of a differential equation are e^{3x} and xe^{3x} ,

then they are linearly independent. Give the general solution of the differential equation. (5 marks)

d) Solve the differential equation $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} = \cos x, y(0) = y'(0) = 1$

(7 marks)

$$\left(\frac{1}{2} + \frac{y^2}{2x}\right) dx = y dy$$

QUESTION THREE (20 MARKS)

a) Solve the differential equation $(x^2 + y^2) dx = 2xy dy$

(6 marks)

b) Show that the differential equation $(ye^{xy} + 2x) dx + (xe^{xy} - 3y + 5) dy = 0$ is exact hence solve it

(6 marks)

c) Solve $(x^2 + 1)y'' + xy' - 4y = 0$

(8 marks)

QUESTION FOUR (20 MARKS)

a) The total cost of production y and the level of output x are related to the marginal cost of production by the equation $(6x^2 + 2y^2) dx - (x^2 + 4xy) dy = 0$. What is the relation

between total cost and output, if $y = 2$ when $x = 1$?

$$\frac{dy}{dx} = \frac{6x^2 + y^2}{2xy}$$

(6 marks)

b) Solve $(1 - x^2) \frac{dy}{dx} - xy = 1$?

(5 marks)

c) A body at a temperature of 50°F is placed outdoors where the temperature is 100°F .

If after 5 minutes the temperature of the body is 60°F , find

i) How long it will take the body to reach a temperature of 75°F .

(6 marks)

ii) The temperature of the body after 20 minutes.

$$t^2(1.2)$$

(3 marks)

$$\frac{11V - v(2v)}{2v} = \frac{11V - 2v^2}{2v}$$

QUESTION FIVE (20 MARKS)

- a) Show that for a RL electric current the amount of current i at any time t is given by

$$i = \frac{3}{R} (1 - e^{-\frac{R}{L}t})$$

Hence sketch the graph showing how the current adjusts from its value of zero to the final steady state value. (10 Marks)

- b) State Newton's law of motion hence determine the differential equation for a freely falling body with air resistance. (3 marks)

- c) Solve $y'' + y = \sec x$ (7 marks)

$$2v^2 - v - 1$$

$$\frac{-2}{-2} \mid \frac{1}{1}$$

$$S = -1$$

$$P = -2$$

$$1 + 2 + 9 - 20 = 0$$

$$8 + 8 + 18 - 20 = 0$$

$$-8 + 8 + 18 - 20 = 0$$

$$-1 + 2 - 9 - 20 = 0$$

$$-64 + 128 + 36 - 20$$

$$64 + 128 -$$

$$-125 + 500 - 45 - 20 = 0$$

$$125 + 50 + 45$$

$$-1000 + 200 + 90 - 20$$

$$-8 + 8 - 18 - 20 = 0$$

$$8 + 8 - 18 - 20$$

$$2v^2 - 2v + v - 1$$

$$2v(v-1) + 1(v-1)$$

$$(2v-1)(v-1)$$

$$2v-1=0$$

$$2v=1$$

$$v=\frac{1}{2}$$

$$\frac{A}{(2v-1)} + \frac{B}{(v-1)}$$

$$A(v-1) + B(2v-1) = 2v$$

$$v=\frac{1}{2}, B=0, A=-2$$

$$B=2$$

$$-\frac{1}{2}A = 2$$

$$A = -4$$

$$= -2$$

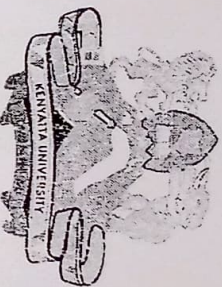
$$2v$$

$$B=2$$

$$\int \frac{2}{2v-1}$$

$$\frac{\partial m}{\partial y} =$$

$$(ye^{xy} + x^2)dx + (xe^{xy} - y^2)dy$$



KENYATTA UNIVERSITY
DEPARTMENT OF MATHEMATICS

ECU202: ODE I CAT II

TIME: 1 HOUR

05/01/2022

Answer **ALL** the Questions.

1. Determine if the functions $f_1 = \sin 2x$ and $f_1 = \cos 2x$ are linearly dependent or independent. (3 marks)
2. Solve the differential equation $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} = \cos x$, $y(0) = y'(0) = 1$ (10 marks)
3. Solve $(D^4 + 2D^3 - 7D^2 + 2D + 70)y = 0$ if one of the roots of the characteristic equation is $-3 + i$, (10 marks)
4. Use the method of undetermined coefficients to solve $y'' - 3y' - 4y = -8e^x \cos 2x$ (7 marks)

1

$$D^4 + 2D^3 - 7D^2 + 2D + 70 = 0$$

13

$$81 + 54 - 63 + 6 + 70 = 0$$

$$81 - 54 - 63 + 6 + 70 = 0$$

$$525 + 250 - 175 + 10 + 70$$

$$1 - 2 - 7 + -2 + 70 = 0$$

$$4 - 16 - 14 - 8 + 70$$



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

SECOND SEMESTER EXAMINATION FOR DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL, MECHANICAL AND BIOSYSTEMS ENGINEERING

ECU 202:ENGINEERING MATHEMATICS VII

DIFFERENTIAL EQUATIONS FOR ENGINEERS

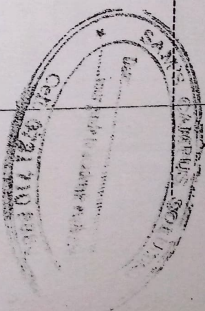
DATE: Tuesday, 2nd July 2019

TIME: 11.00am - 1.00pm

INSTRUCTIONS:

Answer Question ONE and any other TWO questions.

QUESTION ONE (30 MARKS)



- a) Determine the order, degree and linearity of the differential equation $(y'')^{3/2} - 4y' = xy$ (3marks)
- b) Show that x^2 is an integrating factor for exactness for the differential equation $(4x + 3y^2)dx + 2xydy = 0$ hence solve the differential equation (4marks)

- c) Solve the Bernoulli differential equation $\frac{dy}{dx} + y = xy^3$ (5marks)

- d) The initial temperature of a body is 66°C and after 5 minutes its temperature is 56°C . From Newton's law of cooling it is known that the rate of cooling of the body is proportional to the temperature difference between the body and the surrounding medium. Given the room temperature is 34°C , find the equation of the temperature after time t minutes and the time taken for the temperature to drop to 42°C . (5marks)

- e) Use the Method of Undetermined Coefficients to solve the differential equation $y'' - 2y' + 5y = 56\sin 3x + 6\cos 3x$ (6marks)

- f) Use the Variation of Parameters Method to solve the differential equation $y'' + 4y' + 4y = 4x^2e^{-2x}$ (7 marks)

QUESTION TWO (20 MARKS)

- a) Find the general solution to the differential equation $y \tan x \frac{dy}{dx} = (5 + y^2) \sec^2 x$ (4marks)
- b) Determine the particular solution for the initial value problem $y'' - 2y' - 3y = 0$ if $y(0) = 0$ and $y'(0) = -4$ (5marks)
- c) Find the differential equation associated to the primitive equation $xy = Ae^{2x} + Be^{-2x} + x^5$ (4marks)
- d) Use the method of undetermined coefficients to solve the differential equation $(D^2 - 3D - 4)y = 416e^x \cos 2x$ (7marks)

QUESTION THREE (20 MARKS)

- a) Show that the general solution to the differential equation $(x^2 - xy + y^2)dx - xydy = 0$ is $y + x \ln(x - y) = Cx$. (8marks)
- b) Solve the differential equation $(x + y)dx + (3x + 3y - 4)dy = 0$ (6marks)
- c) Solve the differential equation $y'' + 2y' = 3x$ by using reduction of order method (6marks)

QUESTION FOUR (20 MARKS)

- a) Find the orthogonal trajectories to the family of curves $x^2 + y^2 = r^2$ (5marks)
- b) Solve the differential equation $\frac{dy}{dx} = \frac{x + y - 3}{x - y - 7}$ (7marks)
- c) The power series solution to the differential equation $y'' - xy' = 0$ about $x_0 = 1$ is of the form $y = a_0y_1 + a_1y_2$. Find the solution giving up to the terms of power 5 in y_1 and y_2 . (8marks)

QUESTION FIVE (20 MARKS)

- a) A radioactive isotope remains in a laboratory for 10 years after which it is found to contain only 80% of original mass. Find
- i) The half-life of the isotope to the nearest year (4marks)
- ii) The percentage proportion (to the nearest whole number) of the original mass remaining after 25 years (2marks)
- b) The independent solutions to the differential equations $x^3y''' - 9x^2y'' + 36xy' - 60y = 0$ are of the form $y = x^r, r > 0$. Find the values of r and the general solution (7marks)
- c) Use the **Variation of Parameters Method** to solve the differential equation $y'' + y = \tan x$ (7marks)

Table of Laplace Transforms

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$	2. e^{at}	$\frac{1}{s-a}$
3. $t^n, n=1,2,3,\dots$	$\frac{n!}{s^{n+1}}$	4. $t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$	6. $t^{n-1/2}, n=1,2,3,\dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+1/2}}$
7. $\sin(at)$	$\frac{a}{s^2 + a^2}$	8. $\cos(at)$	$\frac{s}{s^2 + a^2}$
9. $t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$	10. $t \cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11. $\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$	12. $\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13. $\cos(at) - at \sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$	14. $\cos(at) + at \sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$
15. $\sin(at+b)$	$\frac{s \sin(b) + a \cos(b)}{s^2 + a^2}$	16. $\cos(at+b)$	$\frac{s \cos(b) - a \sin(b)}{s^2 + a^2}$
17. $\sinh(at)$	$\frac{a}{s^2 - a^2}$	18. $\cosh(at)$	$\frac{s}{s^2 - a^2}$
19. $e^{at} \sin(bt)$	$\frac{b}{(s-a)^2 + b^2}$	20. $e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2 + b^2}$
21. $e^{at} \sinh(bt)$	$\frac{b}{(s-a)^2 - b^2}$	22. $e^{at} \cosh(bt)$	$\frac{s-a}{(s-a)^2 - b^2}$
23. $t^n e^{at}, n=1,2,3,\dots$	$\frac{n!}{(s-a)^{n+1}}$	24. $f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25. $u_c(t) = u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$	26. $\delta(t-c)$ Dirac Delta Function	e^{-cs}
27. $u_c(t) f(t-c)$	$e^{-cs} F(s)$	28. $u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t+c)\}$
29. $e^{ct} f(t)$	$F(s-c)$	30. $t^n f(t), n=1,2,3,\dots$	$(-1)^n F^{(n)}(s)$
31. $\frac{1}{t} f(t)$	$\int_s^\infty F(u) du$	32. $\int_0^t f(v) dv$	$\frac{F(s)}{s}$
33. $\int_0^t f(t-\tau) g(\tau) d\tau$	$F(s) G(s)$	34. $f(t+T) = f(t)$	$\frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$
35. $f'(t)$	$sF(s) - f(0)$	36. $f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
37. $f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$		